



Intelligent Invoice Management

Revenue and Receivables Intelligence

High Level Solution Design

Prototype phase | Version 0.3 | 15 September 2026

This design covers investment-revenue analysis, collection and enforcement monitoring, reconciliation, forecasting, KPI intelligence and conversational reporting. It defines the analytical capabilities and controls to demonstrate in the prototype and the approvals and integrations required for production.

Document Information

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Revision History

Version	Date	Author	Summary of Changes
0.1	24/08/2026	ByteDance	Initial Draft Version
0.2	12/09/2026	ByteDance	Update based on solution owner review 12/09/2026
0.3	14/09/2026	ByteDance	Round-2 review: repositioned as revenue and receivables intelligence; corrected Sanad/Efaa/Makeen/Furas platform descriptions; added Sanad Enforcement-to-Invoice Mapping (WF-09); expanded KPI & Performance Intelligence and Dynamic Report Generation; added governed semantic/business metrics layer; removed remaining procurement/tax-compliance remnants from text and diagrams; strengthened data-authority matrix; aligned acceptance criteria and benefit statements with clarified scope

HLSD Review Checklist

Review responsibilities below are carried forward from the source template. No review or approval is represented as completed by this revision.

Review item	Sections	Required review roles
Business problem and strategy	2.1–2.2	OT / SO / AC / CAIT
Source landscape and data authority	2.5 / 3.1	OT / SO / data owners / CAIT
Requirements, acceptance and benefits	4–5	OT / SO / AC / CAIT
Executive summary	Executive Summary	OT / CAIT / SGG / SH / CAIC
Personas, workflows and journeys	3.2–3.4	OT / SO / CAIT
Roadmap and open dependencies	6	OT / SO / AC / CAIT
Architecture	3.1 / 4.2	NHCI ARB technical review; no prototype approval implied
Stakeholders	2.3	OT / SO / AC / CAIT

OT = Oversight Team; SO = Solution Owner; AC = AI Champion; CAIT = Central AI Team; SGG = AI Steering Group; SH = Sector Head; CAIC = AI Central Committee; ARB = Architecture Review Board.

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Section entries below are refreshed against the rendered revised document. Workflow and requirement IDs remain stable for review traceability.

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Executive Summary

Intelligent Invoice Management is a revenue and receivables intelligence solution for municipal investment and lease revenues, collections and enforcement oversight. Existing platforms retain transactional ownership; this solution maintains analytical mappings, evidence, forecasts and reports. Tahseel remains authoritative for official collection figures, while other attributes follow the field-specific authority matrix in Section 3.1.1.

The priority business outcome is a reliable link from Sanad non-fine enforcement records to their underlying SADAD invoices. The platform checks structured references first and retrieves associated documents only when links are unusable. It separates extracted numbers from validated relationships, supports one case with several invoices and preserves unsupported cases as unresolved without duplicating case values or invoice balances.

A governed semantic metrics service supports KPI and Performance Intelligence, forecasting and dynamic conversational reporting. Users can create and refine editable reports, save definitions and refresh or distribute them within approved permissions. Official KPI definitions, baselines, targets and thresholds change only through authorized approval and versioning; historical commitments remain reproducible.

Relevant Furas-originated investment/lease data is in scope. Procurement, supplier payments, tax compliance, contract-award/amendment administration, direct payment processing and platform-executed legal actions are excluded. Human review remains essential for ambiguous mappings, governance changes and report distribution.

The prototype uses controlled sample data and simulated integrations. Production depends on owner-approved source roles, field authority, formulas, access, validation data and security design. The supplied files do not establish live integration availability, an approved KPI catalog or completed stage-gate approval.

Business benefits will be measured as released employee capacity, collection outcomes attributable to the solution and adoption. Previous revenue-accuracy and GDP-linked growth claims are not accepted commitments. Section 5 defines measurement and approval requirements; Section 6 retains the original milestone dates as proposed dates subject to readiness and re-baselining.

1 Introduction

1.1 Purpose

This Prototype Phase HLSD establishes the solution scope, architecture, personas, workflows, requirements, acceptance criteria and delivery dependencies. It supports review of the proposed revenue-intelligence design and planning of subsequent delivery phases; it does not certify implementation or approve official measures.

1.2 Scope

Prototype demonstrations use sample or simulated data. The following capabilities describe the target solution; production integrations and official reporting definitions are subject to the gates in Section 6.3.

In scope

- Controlled intake of approved invoice, collection, investment/lease and enforcement datasets with source lineage and normalization.
- Sanad enforcement-to-invoice mapping: structured references first, all relevant SADAD numbers from document text/OCR where required, evidence-backed validation and one-to-many relationships.
- Duplicate and amount-consistency checks, cross-platform reconciliation, revenue anomaly review and risk-based case routing under confirmed revenue responsibilities.
- Collection forecasting, KPI and Performance Intelligence, editable conversational reports, periodic reports and controlled export/distribution.
- Relevant municipal investment and lease-contract data originating from Furas are in scope for revenue and receivables analysis. Procurement, supplier-payment processing, and the execution or administration of contract-award and amendment processes remain outside this solution's scope.
- Existing tax/VAT values may be retained as optional source attributes where needed to interpret or reconcile amounts. VAT extraction is not mandatory for Sanad mapping.

Out of scope

- Procurement, supplier-payment approval/processing, contract-award/amendment execution or administration, and general contract lifecycle management.
- Direct payment processing, bank-account reconciliation, replacement of source platforms, or silent changes to official records.
- Tax assessment, tax-registration verification, VAT/tax compliance and any ZATCA or other tax-authority integration.
- Initiating, pausing, cancelling or amending legal enforcement actions, municipal receivables or violations in this platform.
- Entity relationship-management operations and issuance of legally approved official financial reports. Analytical entity relationships required for revenue measures remain in scope.

Use the precise term “SADAD invoice number” for the invoice identifier. A payment-order field may be introduced only if its data owner confirms a relevant revenue-collection meaning; no such confirmation was supplied.

1.3 Intended Audience

Audience	Purpose of Review
Central AI Committee	Strategic review of an executive version of the HLSD and final stage-gate approval
AI Solutions Steering Group	Strategic and technical review of an executive version of the HLSD, stage-gate endorsement for moonshot solutions and stage-gate approval for standalone solutions

Audience	Purpose of Review
Sector Head	Review for alignment with sector needs and priorities, provide final approval before submission to Steering Group / Central AI Committee
AI Champion	Review for sector alignment
NHCI ARB	Review for technical architecture (no approval at this stage)
NHCI Technical Lead(s)	Day-to-day delivery management and review
Oversight team	Governance review, standards compliance, and delivery assurance across the program
AI Solution Owner	Solution-level owner; provide input, review key requirements, business outcomes, and user journeys
Solution Vendor	Document owner - produced and maintained by vendor. Responsible for completeness and accuracy of all sections.

Governing-body review also requires an executive summary presentation covering the HLSD, per the source template.

1.4 Definitions and Acronyms

Term / Acronym	Definition
HLSD	High-Level Solution Design
NHCI	NHC Innovation
MOMAH	Ministry of Municipalities and Housing
ARB	Architecture Review Board
AI	Artificial Intelligence
Preparation phase	Define the vision, objectives, scope, requirements, and high-level solution approach
Prototype phase	Validate UX/UI, key features, and user journeys before full development
Alpha phase	Build and test core features in a sandbox environment to prove the technical approach
Beta phase	Complete feature development and validate the solution with integrations in a staging environment
Release & Stabilization phase	Deploy to production, release improvements, fix issues, and stabilize the solution
Operate phase	Hand over to operations/technical teams and drive adoption through change management and ongoing support
BRD	Business Requirements Document; defines the scope, business needs, functional requirements, and constraints for the Intelligent Invoice Management solution.
Human-in-the-Loop (HITL)	Human governance for ambiguous/critical analytical decisions, KPI changes and report distribution. Routine approved calculations may run automatically within authorized rules.
Optical Character Recognition (OCR)	Technology that converts scanned or image-based documents into machine-readable text, enabling automated extraction and analysis of invoice data.

Term / Acronym	Definition
Personal Data Protection Law (PDPL)	Saudi Personal Data Protection Law issued by SDAIA; the primary data-protection regulation applicable to this solution.
Service Level Agreement (SLA)	A formal agreement defining guaranteed performance levels such as availability, response time, and service quality between the Ministry and the implementation partner.
Role-Based Access Control (RBAC)	An access-control model that restricts platform features and data according to user role and the need-to-know principle.
Tahseel Platform	Government payment platform of the Ministry of Municipalities and Housing for collecting revenue from municipal services and government payments.
Makeen Platform	Working description: candidate billing/invoice-record source with relevant revenue and possible investment/lease references. Exact coverage, Furas-originated data availability and ownership require Makeen-owner confirmation; no existing integration is assumed.
Efaa Platform	Working description for WF-08: candidate violations/penalty data source. Its exact domain, enforcement-referral role and relationship with Mumtathil require data-owner validation. It is not treated here as a contract/payment-order platform.
Sanad Platform	Platform supporting municipalities in pursuing legal recovery of outstanding investment debts and other non-fine, non-violation municipal receivables through the Board of Grievances; distinct from the Efaa fines and violations workflow. Its enforcement/case reference and Board of Grievances reference are distinct from, and not always linked to, the underlying SADAD invoice number(s).
Furas Platform	Enables municipalities to offer municipal real estate to investors through competitive bidding, resulting primarily in investment and lease contracts that generate municipal receivables. Relevant contract data is in scope for revenue and receivables analysis, invoice matching, collection monitoring, forecasting and enforcement analysis; the required data feed and whether it is obtained directly or through an existing integration or approved dataset is to be confirmed with the Furas data owner.
Mumtathil Platform	Working description for WF-08: candidate municipal violation/case-status source. Judgment, appeal and enforcement field availability and authority remain pending owner confirmation.
Confidence Score	Model- or rule-specific confidence indicator, with method/version and calibration limits. Extraction confidence and relationship validation are separate assessments; neither alone constitutes legal or business approval.
AI Agent	A specialized AI program that independently performs a specific task within defined boundaries, such as the Ingestion Agent or Validation Agent.
Financial Forecasting Model	Model estimating collection or risk outcomes for a stated horizon using approved features; available outcomes depend on labeled data. Predictions are distinct from official source status.
Financial Anomaly	Any invoice, item, or pattern that deviates materially from normal expectations based on historical behavior models.
Government ERP	Government Enterprise Resource Planning system; the integrated environment for managing financial, human, and operational resources in Saudi government entities.
Audit Trail	An immutable sequential record documenting every action in the system: who performed it, when, what changed, and for what reason.
National Cybersecurity Authority (NCA)	Government authority responsible for cybersecurity in the Kingdom; it issues controls and frameworks with which this solution must comply.

Term / Acronym	Definition
SADAD invoice number	Invoice reference retained as a string; distinct from Sanad record ID, Board of Grievances enforcement reference and internal analytical record ID.
Semantic metrics layer	Governed entities, relationships, reusable measures, source authority, lineage, permissions and effective-dated business definitions.
Unresolved mapping	A proposed case–invoice relationship without sufficient validation evidence; it is not assigned an inferred invoice number.

2 Business Context

2.1 Business Problem

Revenue information is distributed across Tahseel, Makeen, Sanad and other source platforms. Inconsistent identifiers and fragmented access make reconciliation, investment-revenue analysis and enforcement monitoring difficult. A Sanad enforcement reference may lack a usable structured link to one or more SADAD invoices, limiting reliable portfolio reporting.

The solution addresses intake and normalization, evidence-backed validation, anomaly review, case routing and governed analytics. Descriptions of current platform capabilities and manual cycle times in the source are working assumptions, not verified platform assessments.

#	Pain Point	Business Impact	Severity / Impact
1	Invoice data is fragmented across multiple platforms (Tahseel, Makeen, Efaa, Sanad)	Data duplication and inconsistency and difficulty obtaining a unified view	High
2	Reliance on manual invoice processing and reconciliation	Higher operating cost, slow processing cycle, and more human errors	High
3	No automated validation of invoice matching across different platforms	Risk of uncollected or misapplied revenue due to unmatched or incorrect invoices	High
4	Difficulty detecting duplicate invoices and potential manipulation	Potential revenue leakage and unreliable portfolio totals; anomaly evidence requires investigation.	High
5	Limited ability to predict collection or cancellation probability	Difficulty in financial planning and reduced accuracy of revenue forecasts	High
6	Incoming invoice formats are not standardized across platforms	Intensive manual effort to standardize data before processing	High
7	No consolidated view with agreed data freshness	Delayed visibility and inconsistent management information	High
8	Difficulty routing invoices automatically to the responsible reviewing officer	Delayed case review and accumulation of unmonitored invoices	High
9	Limited transparency in financial AI decisions	Difficulty in auditing and accountability in official financial reporting	High
10	Difficulty connecting legacy financial systems with modern digital platforms	Technical barriers limiting integration and end-to-end automation	High

Opportunity statement

Automated matching, controlled exception handling and reusable revenue measures can reduce reconciliation effort and improve follow-up prioritization. The source's 15–30% revenue-accuracy improvement lacks a defined denominator and supporting baseline, so it is withdrawn as a quantified commitment. Section 5 requires measured baselines and an attribution method before benefit targets are approved.

2.2 Strategic Alignment

Alignment area	Design contribution
Vision 2030 Objective	Improve government performance efficiency, non-oil revenue oversight and transparent revenue management.
NHCI Strategic Objective	Aligned with the core objectives of NHCI's AI innovation implementation and digital transformation of government services, aiming to develop a reusable benchmark solution for government AI agents.
MOMAH Sector Strategic Objectives (SSOs)	Support municipal financial sustainability and revenue oversight. Source v0.3 cites 85% financial sustainability and SAR 25B invoicing by 2025; these historical strategic references require owner confirmation of relevance to the 2026 plan and are not project acceptance targets.
MOMAH AI Strategy Alignment	Apply governed AI extraction, relationship validation and forecasting with evidence and human oversight for material analytical decisions; source-system authority remains unchanged.

2.3 Key Stakeholders

Stakeholder and role	Named contacts from source	Responsibility
AI Solution Owner — Solution-level owner	Tariq Aljarboa, Abdulaziz Alajlan taljarboa@momah.gov.sa, aa.alajlan@momah.gov.sa	Participate in vendor check-ins, Sprint Reviews, Steering Group, and design discussions
AI Champion — Sector-level coordinator	Waleed Al-Askar W.Alaskr@momah.gov.sa	Participate in Sprint Reviews and Bi-Weekly Sector meetings
Sector Head — Sector-level sponsor	Omar Almhasen Oalmhasen@momah.gov.sa	Participate in Bi-Weekly Sector meetings and Central AI Committee sessions

Additional accountable roles are the source data owners, Business KPI Owner, Data Governance Lead, Delivery Lead and IT/Security/QA leads. Individuals and acceptance delegates must be assigned before the relevant gate; role assignment here does not imply their approval.

2.4 Expected Business Outcomes

- Reliable investment-receivable and enforcement analysis without double-counting; traceable unresolved coverage rather than forced matches.
- Faster review and reconciliation, measured through comparable time studies. The source's "up to 50%" processing-time reduction is an unvalidated hypothesis, not an approved commitment.
- Evidence-supported collection follow-up, forecast monitoring and management reporting using consistent approved definitions.
- Improved capacity and adoption, with cash savings reported only when separately evidenced; source platforms remain authoritative.

2.5 Current Solution Landscape

Source / role	Relevant use and confirmation needed
Tahseel	Official collection figures; field dictionary, event/reversal semantics, access and update frequency require confirmation.
Makeen	Proposed billing and revenue reference data; confirm scope, identifiers and field authority.
Sanad	Non-fine/non-violation municipal receivable enforcement records and associated evidence; confirm access and case/identifier semantics.
Furas	Investment/lease contract origin; confirm direct, verified intermediary or approved dataset route.
Efaa / Mumtathil	Working violation/case-status sources for WF-08; both owners must validate roles and data authority.
Other government/finance datasets	Only approved revenue-relevant datasets; optional expense and master data need a confirmed owner and access agreement.

The supplied files do not substantiate platform launch dates, existing API availability, budgets, absence of AI features or a 3–5 day baseline. Those prior assertions are not used as verified facts in this design.

3 Solution Overview

3.1 Solution Description

The proposed platform consolidates authorized read-only revenue data while preserving source identity, document evidence and field-level lineage. Five logical capabilities support intake, validation, anomaly detection, review routing, and analytics/forecasting. These are logical services; implementation does not require five independently autonomous agents.

Figure 3.1-1 Logical architecture

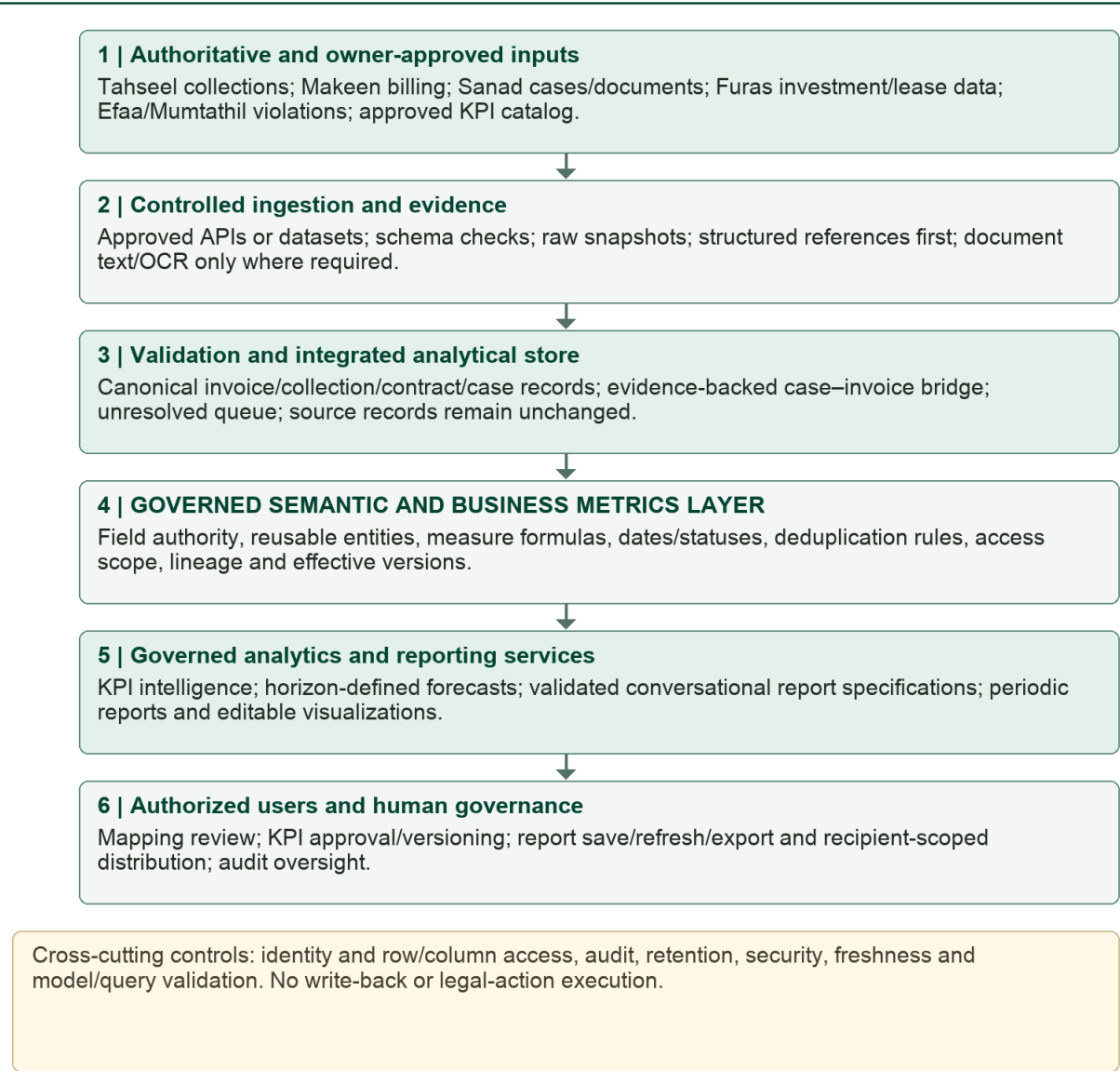


Figure 3.1-1 Logical architecture

3.1.1 Source Authority and Integration Boundaries

This is a proposed authority matrix for owner agreement. The feedback confirms Tahseel's role for official collection figures; it does not approve every field assignment below. Retain conflicting source values and provenance. An unresolved ownership decision must not be resolved by an arbitrary source override.

Domain and material fields	Authority and current status	Owner confirmation and handling
Official collections: collected amount, collection event/date and relevant collection status	Tahseel for official collection figures, as directed in feedback. Exact field definitions, reversals and cutoff remain to be agreed.	Tahseel data owner + revenue owner confirm data dictionary and collection methodology; retain lineage and disclose differences.
Billing: native invoice key, SADAD link, issued amount, billing status, due date, product/revenue classification	Proposed: Makeen or the confirmed originating billing system. Not approved by the supplied sources.	Makeen/billing owner approves fields and mappings; do not overwrite them merely because Tahseel differs.
Investment and lease contract: ID/version, dates, lease terms, billing schedule, asset and investor reference	Furas-originated contract record, accessed through an owner-confirmed route. No live integration established by the supplied files.	Furas owner approves originating fields, lawful access, feed route and crosswalk to billing; discrepancies remain visible.
Non-fine enforcement: Sanad record ID, Board of Grievances reference, case amount, status and source documents	Sanad-provided records/evidence; legal reference/status source and update semantics require Sanad-owner confirmation.	Sanad owner validates identifiers, evidence association and legal status lineage; no legal-state inference by the platform.
Violations, penalties, appeals, judgments and enforcement states	Efaa and Mumtathil roles and field authority are provisional pending both owners' validation.	Efaa/Mumtathil owners resolve the domain dictionary and event model; conflicting or unsupported state remains unresolved.
Official KPIs, baselines, targets, improvement directions and business thresholds	Ministry-approved business KPI catalog and responsible business owner; actual catalog/sign-off not supplied.	KPI owner approves formulas, dates and baseline method. Prototype examples stay marked illustrative.
Organization hierarchy, Amanah/municipality and revenue-source codes	Proposed Ministry master-data catalog; specific owner and dataset to be confirmed.	Data Governance Lead appoints master-data owners and approves effective-dated crosswalks.
Optional operating-expense values, coverage methodology and target periods	Approved finance dataset and finance/business owner, not procurement transactions; not supplied.	Do not calculate coverage until numerator, denominator, access and aligned periods are approved.
Analytical links, candidate matches, review decisions, model/report definitions and recommendations	This platform owns its traceable analytical artifacts only.	Named reviewer/definition owner approves as applicable. Analytical state never overwrites source records.

Furas minimum dataset and route

Request originating contract ID/version and contract type (investment/lease), investor reference, asset/property reference, Amanah/municipality, start/end dates, billing schedule and due dates, rent/receivable amounts and currency, status and relevant amendment-effective dates, available invoice/SADAD crosswalks, and source update/evidence references. Guarantee attributes are optional only when relevant and owner-approved. Consume terms as revenue data; do not administer awards or amendments.

The Furas owner must choose and evidence one route: direct approved feed, a verified existing integration (with the intermediary and lineage identified), or a controlled approved dataset. Confirm field completeness, permissible use, cadence and missing-data behavior before related Alpha design is

baselined. Where unavailable, the prototype uses labeled samples and production analysis discloses the gap.

3.1.2 Semantic and Business Metrics Contract

Governed definition	Required approved content and release rule
Entity grains and keys	Invoice fact: unique source-native invoice ID plus approved SADAD crosswalk; collection fact: unique collection transaction/event with reversals; contract: originating contract ID/version; enforcement case: Sanad record ID and separate Board of Grievances reference. Preserve unresolved identities.
Relationships and allocation	Use a versioned case–invoice bridge carrying evidence and validation status. Sum invoice balances once per invoice and case amounts once per case. Aggregate each grain before combining views. If a source later links one invoice to multiple cases, prevent repeated balances across those cases. Allocate case amounts only under an approved method/version; otherwise show unallocated amounts separately.
Net invoicing	Define $NI(P,C,v) = \text{SUM of owner-approved signed invoice events/adjustments included by rule version } v \text{ for period } P \text{ and population } C$. Approval must specify included statuses, cancellations, credit notes, reversals, exclusions, relevant date, currency and tax treatment. This is a configuration contract, not a claim that the Ministry’s formula has been supplied.
Collection rate	Define $CR(P,C,v) = 100 \times NI(P,C,v) / D(P,C,v)$, only when the approved denominator is nonzero. Owner must specify numerator event/status and reversal treatment, denominator (including whether approved net invoicing or another eligible receivable base), exclusions, cohort, calendar and date alignment. Zero/missing denominator returns “not applicable”/“unavailable”, not an invented percentage.
Time and status semantics	Distinguish invoice issue date, due date, collection event/value date, source update time, ingestion time, enforcement referral date and snapshot as-of date. Maintain source-to-business status mappings. Report source cutoff and organizational/revenue-classification versions.
KPI catalog contract	Each KPI stores ID/name, rationale, formula/version, unit, owner, measurement period, dimensions, aggregation/exclusion rules, baseline period/population/method/value, target, intended direction/range, threshold, approver and effective dates. Proposed records cannot silently become official.
Lineage and authorization	Every result links to source snapshot and governed measure version. Scope queries and evidence by owner-approved role/row/column policies. Preserve historical commitments and govern any restated historical series separately.
Release gate	The actual approved net-invoicing and collection-rate methods are not included in the two supplied files. Prototype formulas must be marked illustrative; production official KPI use is blocked until the business owner approves populated definitions and reconciliation examples.

Reporting partitions must be explicit: “referred” invoices are distinct invoices with at least one validated, relevant case relationship; “not referred” requires adequate confirmed referral coverage; unknown or unresolved referral status is separate. Collections on referred invoices are not automatically collections attributable to enforcement. Use an owner-approved timing/attribution rule for that measure.

3.1.3 Conversational Reporting Technical Approach

Proposed approach: an Arabic conversational interface produces a schema-validated report specification containing approved measure IDs/versions, dimensions, filters, period, comparisons, calculations and visualization layout. A policy service validates scope and allowed operations. A semantic query service compiles parameterized read-only queries; a report service renders an editable report and stores its versioned definition. The language model has no direct source-database credentials and cannot invent production formulas.

Refinements modify the report specification and repeat validation. Saved definitions pin measure versions; scheduled refresh rechecks the owner and each recipient, identifies incompatible/retired definitions and requires controlled migration. PDF and XLSX outputs carry source, period, filters, calculations, version and freshness metadata. Restricted content is omitted through scoped copies or delivery is denied.

Power BI describes the intended analyst-like experience; it is not a prescribed product dependency. OCI hosting is the source's proposed baseline, pending architecture/security validation. Specific model, OCR, query, report-rendering and identity services are not selected in the supplied files. Approval of Saudi-hosted endpoints, network connectivity, licenses, document storage, access policies and operational support is required before real-data deployment.

Freshness is source-specific: the original five-minute dashboard claim conflicts with daily, weekly and monthly feeds. Each dataset requires an agreed cadence and SLA; every display shows source cutoff and stale-data state. Forecasts and reports cannot be fresher than their available inputs.

3.2 User Personas

Persona	Role and primary need	Interaction / frequency
Invoice and Collection Officer	Financial employee responsible for daily invoice ingestion and validation. Process invoices quickly and accurately and detect errors and duplicates	Uses the invoice and daily dashboard to review ingested invoices and alerts Daily
Revenue and Collection Manager	Manager responsible for monitoring collection performance and making intervention decisions. Source-aware view of collections, forecasts and deviations	Reviews evidence, prioritizes follow-up and approves changes only within confirmed revenue responsibilities. Weekly / Monthly
Internal Financial Auditor	Auditor of revenue data quality and analytical decisions. Verify mapping evidence, review permissions and traceability	Reviews analytical changes, audit history and governance exceptions; no supplier-payment approval path. Periodic / As needed
Shared Services Leader	Executive leader responsible for performance of the invoice and revenue ecosystem. Comprehensive executive view of performance, risks, and financial impact	Reviews summary reports and strategic indicator dashboards Monthly
Governance Officer	Owner of analytical review/access governance. Verify authorized reviewers, KPI changes and report distribution	Reviews revenue responsibility rules, escalations and authorization deviations. Periodic / As needed
Financial Reconciliation Officer	Financial officer responsible for reviewing billing differences between Makeen and Tahseel.. Reconcile differences by field authority; Tahseel governs official collection figures only.	Reviews differences and proposed corrections, confirms analytical findings and refers source changes to the authorized owner. Weekly
Penalties and Fines Officer	Officer responsible for monitoring penalties and fines and their appeal and enforcement status across Efaa and Mumtathil.. Track violation status, enforcement progress, appeals, and their impact on collection probability.	Reviews owner-confirmed source states and follow-up recommendations; legal actions occur only in existing source processes. Daily / As needed
Field Collection Manager	Collection execution role responsible for acting on collection-priority recommendations generated from forecasting results.. Prioritize follow-up activities and focus collection effort on invoices requiring proactive intervention.	Uses collection-risk rankings, collection/delay/cancellation probabilities, and intervention recommendations to execute follow-up actions on prioritized invoices. Daily

Persona	Role and primary need	Interaction / frequency
Amanah Leader	Amanah-level management role receiving revenue-performance information for oversight and performance review.. Review revenue/collection performance; optional expense coverage only with approved data and formula.	Reviews the periodic revenue report/dashboard, including collected-versus-target performance, revenue-source analysis, and operating-expense coverage at Amanah level. Monthly

3.3 Processes and Workflows

All nine target workflows describe the intended analytical behavior. Prototype feeds, scores, external notifications and source responses are simulated and visibly labeled. Internal review and versioning may be demonstrated; there are no official source writes. Routine calculations run within approved rules; exceptions and governance changes require authorized people.

Workflow	Capability	Related requirements
WF-01	Intelligent Invoice Processing	FR-001–009/011
WF-02	Collection Forecast and Follow-up Prioritization	FR-010/020
WF-03	KPI and Performance Intelligence	FR-010/017/019
WF-04	Conversational Reporting and Dynamic Report Generation	FR-018/019
WF-05	Financial Audit Review	FR-007/012
WF-06	Periodic Revenue Reporting	FR-013/018/019
WF-07	Financial Reconciliation	FR-005/014/019
WF-08	Violation Follow-up	FR-015
WF-09	Sanad Enforcement-to-Invoice Mapping	FR-001/002/005/016/019

Workflow IDs are the authoritative identifiers in this revision. Legacy UC-01–UC-13 and SCR-05/SCR-09 references in source v0.3 must be cross-checked in DOC-01 and DOC-02 before submission; those companion artifacts were not supplied.

WF-01 Intelligent Invoice Processing

Current-state context: Invoice and receivables records are reviewed across separate platforms; intake, reconciliation and exception coordination require manual consolidation.

Target-state steps

Step	Activity and outcome	Responsible role
WF01-01	Receive a record through authenticated manual upload or an approved source intake; assign a unique analytical record ID and retain source identifiers.	Invoice and Collection Officer
WF01-02	Read structured fields first, then machine-readable text or OCR when needed. Keep original values, normalized values and evidence separately. Route Sanad linkage gaps to WF-09.	Ingestion Agent
WF01-03	Check completeness and duplicates using source keys and supporting evidence. Quarantine unsupported or suspected duplicate analytical records for review; preserve the original receivable.	Ingestion Agent
WF01-04	Validate SADAD references and relevant investment-receivable data against approved datasets and field authority. Record relationship validation separately from extraction confidence.	Validation Agent
WF01-05	In parallel, assess amount consistency and unusual revenue patterns; preserve evidence and flag risk for review. Tax compliance and received-service validation are excluded.	Anomaly Detection Agent
WF01-06	Join both check results. Clear records continue to analytical monitoring; unresolved or conflicting results enter the exception queue.	Platform
WF01-07	Review the proposed extraction, classification or mapping; confirm, reject the proposal, request evidence or escalate. No decision cancels an invoice or municipal receivable.	Internal Financial Auditor
WF01-08	Route review cases using approved revenue responsibilities, organization and risk rules; send an internal notification with evidence and confidence.	Routing Agent
WF01-09	Review notified cases and plan follow-up under WF-02. Mandatory human review remains for ambiguous mappings, KPI changes and report-distribution authorization.	Revenue and Collection Manager
WF01-10	Record analytical status and review events. Display official collection status only from the authorized source snapshot; do not update that source.	Platform
WF01-11	Correct a derived proposal, retrieve fresh source data, request documents or escalate. Reprocessing repeats checks; rejected proposals remain auditable and do not delete source records.	Officer / Auditor

Figure 3.3-1 WF-01 target workflow



Figure 3.3-1 WF-01 target workflow

WF-02 Collection Forecast and Follow-up Prioritization

Current-state context: Follow-up depends on source reports and manual prioritization; the availability of historical training data and predictive capability must be confirmed.

Target-state steps

Step	Activity and outcome	Responsible role
WF02-01	Run the proposed daily cycle before 08:00 Asia/Riyadh, subject to feed availability; display the actual source cutoff and any delay.	Platform
WF02-02	Use approved semantic-layer features: invoice balances, due dates, collection history and Furas-originated investment/lease terms when available. Guarantees or market indicators require owner-approved access and definitions.	Analytics Agent
WF02-03	Estimate collection by the approved horizon, overdue risk and cancellation likelihood only where labels support it; publish horizon, model version, data cutoff and uncertainty. Scores are not official statuses.	Analytics Agent
WF02-04	Apply approved revenue risk and follow-up thresholds. Rank invoices without duplicating balances through enforcement joins. Unsupported or cold-start cases carry an insufficient-history flag.	Analytics Agent
WF02-05	Update SCR-09 with scores, forecast horizon, priority, evidence and freshness. Use approved fallback models only with disclosed limitations; otherwise show unavailable.	Platform
WF02-06	Notify the responsible manager of high-risk cases under the approved alert policy; do not imply a statutory deadline unless confirmed by the source owner.	Platform
WF02-07	Review the ranked portfolio and approve operational follow-up priorities within existing revenue responsibilities.	Revenue and Collection Manager
WF02-08	Perform approved follow-up in existing collection processes; the analytical platform neither executes collection nor changes legal or receivable status.	Field Collection Manager

Figure 3.3-2 WF-02 target workflow

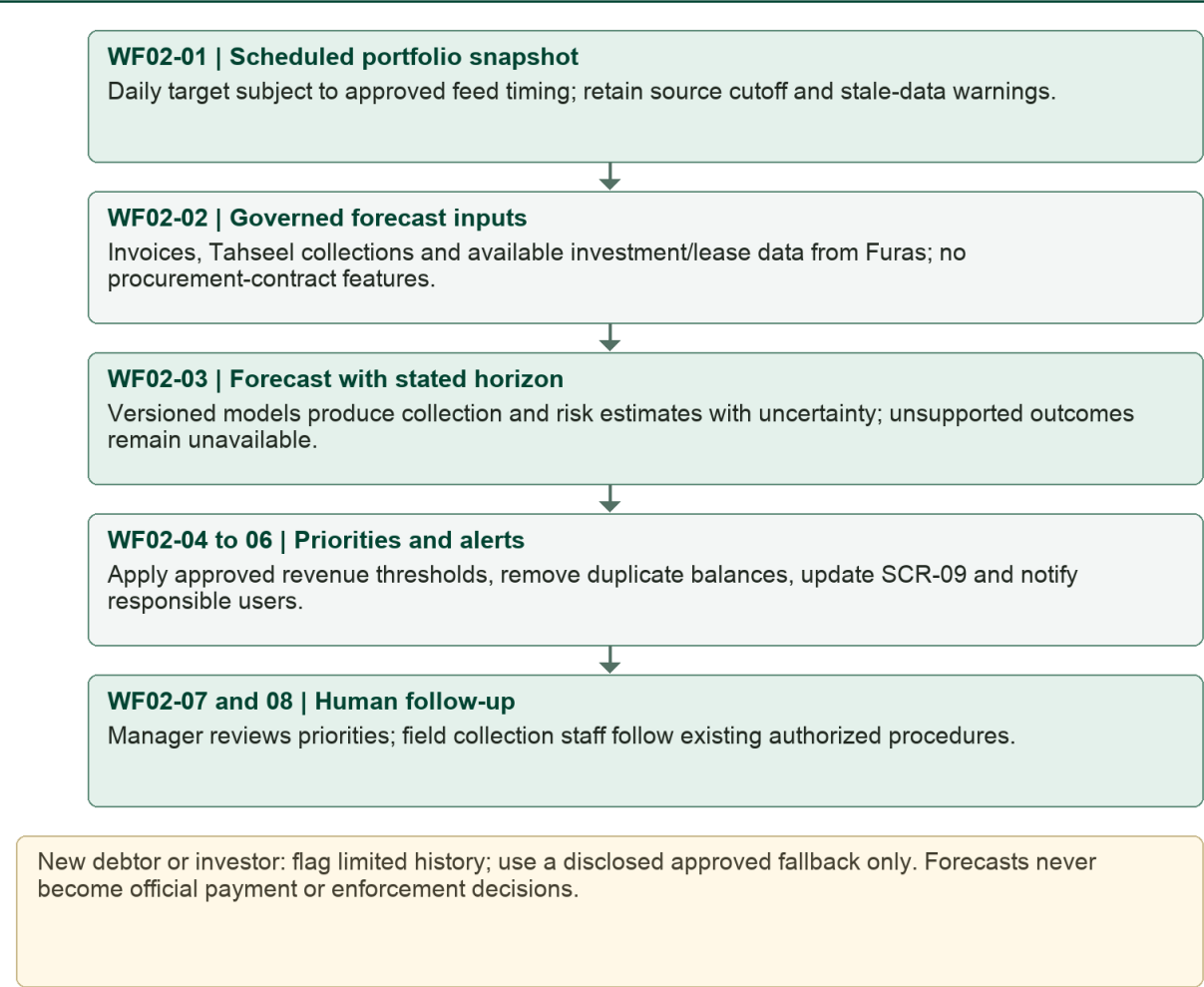


Figure 3.3-2 WF-02 target workflow

WF-03 KPI and Performance Intelligence

Current-state context: Performance is consolidated from platform reports and spreadsheets; official KPI definitions and historic approvals must be obtained from business owners.

Target-state steps

Step	Activity and outcome	Responsible role
WF03-01	Authenticate and open SCR-05 with the required organizational and revenue access scope.	Manager / Shared Services Leader
WF03-02	Load the approved KPI catalog and version applicable to the measurement period; show each source cutoff. Freshness follows the approved feed SLA, not an unconditional five-minute promise.	Platform
WF03-03	Retrieve actuals, approved forecasts and anomaly evidence through the semantic layer; do not infer causation from correlation.	Analytics Agent
WF03-04	Show the approved KPI set with formula, owner, period, baseline, target and intended improvement direction. The former six indicators are candidates, not a mandatory or approved catalog.	Platform
WF03-05	Drill down by authorized organization, revenue source and period; retain the measure version and comparable population.	Manager / Shared Services Leader
WF03-06	Evaluate adverse deviation using each KPI's increase/decrease/range direction and approved thresholds, rather than treating every below-target value as poor performance.	Platform
WF03-07	Create or refine an editable management report through WF-04 using the same governed definitions.	Manager / Shared Services Leader
WF03-08	Save, export or schedule the report only after recipient authorization checks; preserve report definition, source cutoffs and measure versions.	Platform
WF03-09	Identify trends, material deviations and evidence-supported contributing factors; distinguish associations from proven causes and recommend follow-up.	Analytics Agent
WF03-10	Propose additional KPIs with rationale, formula, input data and decision value. Proposed baselines must specify historical period, population and method.	Analytics Agent
WF03-11	Recalibrate analytical expectations within approved rules. Official KPI, baseline, target or threshold changes require business-owner approval, version and effective date; retain the prior period's commitments.	Business KPI Owner

Figure 3.3-3 WF-03 target workflow



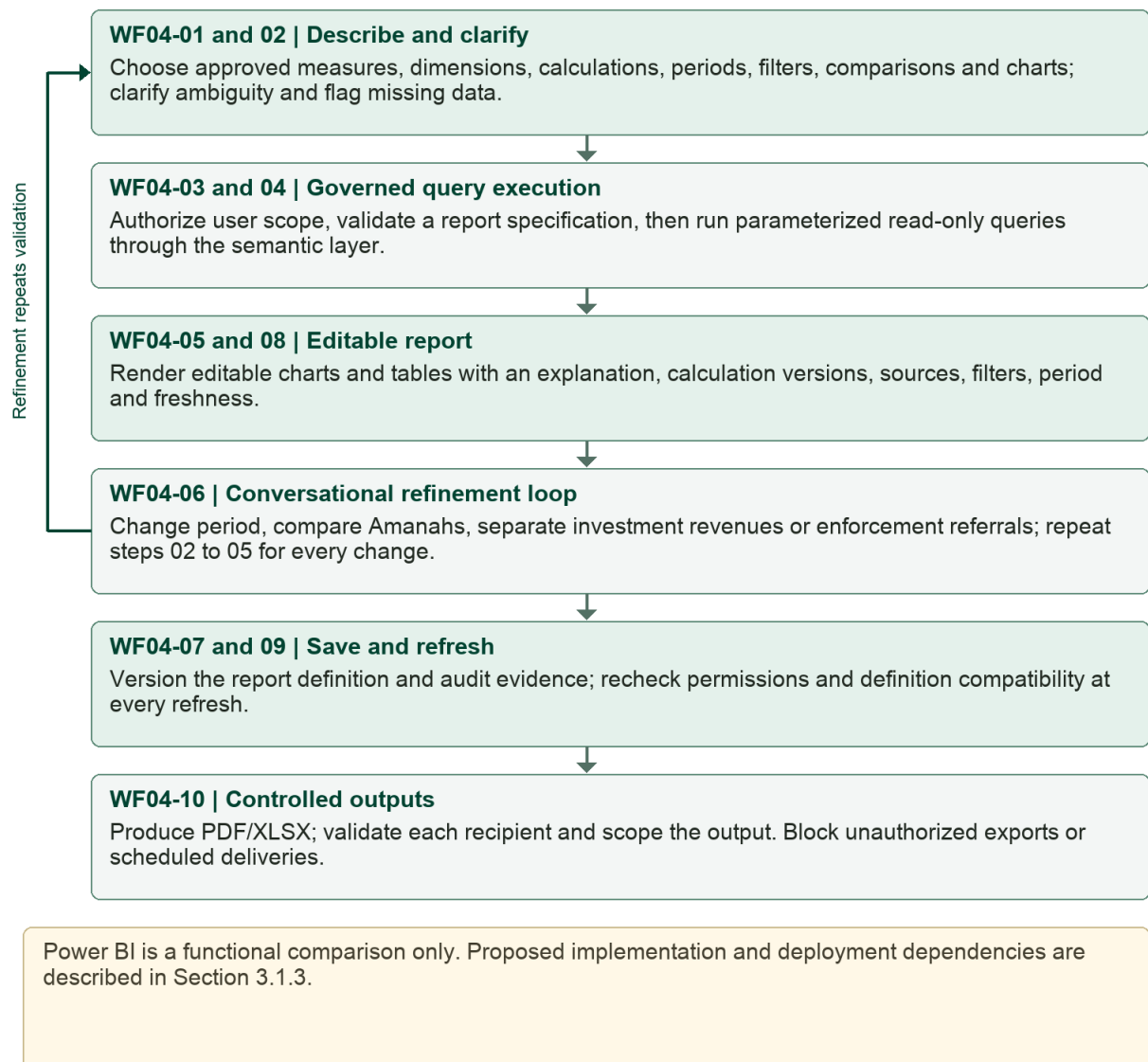
Figure 3.3-3 WF-03 target workflow

WF-04 Conversational Reporting and Dynamic Report Generation

Current-state context: Users depend on fixed dashboards and manual analysis; the target is an analyst-like reporting experience through an authorized conversational interface.

Target-state steps

Step	Activity and outcome	Responsible role
WF04-01	Describe an analysis or ask a question in Arabic; a report request can specify investment revenues, period, Amanah comparison or enforcement referral.	Authorized Business User
WF04-02	Resolve intent, approved measures, dimensions, periods, filters, comparisons, calculations and visualizations. Ask for clarification when ambiguous; list unavailable data without invented values.	Analytics Agent
WF04-03	Apply identity, role, organization, row/column permissions and report-distribution rights before query planning. Deny unauthorized data requests.	Platform
WF04-04	Compile a validated report specification into parameterized, read-only queries through the governed semantic service. Reject unapproved formulas and direct raw-table access.	Authorized Data Service
WF04-05	Return a grounded answer or editable report with charts/tables and explanation; include sources, period, filters, calculation definitions/versions and data freshness.	Analytics Agent
WF04-06	Refine period, Amanah comparison, investment-revenue separation or enforcement classification conversationally. Each edit repeats clarification, authorization and calculation checks.	User / Platform
WF04-07	Log user, report specification/version, authorized query, measure versions, source snapshot, result reference and timestamp under approved retention and access rules.	Audit Service
WF04-08	Allow users to edit report layout, selected measures and approved calculations in the report builder; do not require a separately developed dashboard for each report.	User / Report Service
WF04-09	Save a versioned report definition; on refresh, revalidate owner permissions, source availability and measure compatibility. Suspend unauthorized jobs and disclose definition changes.	Report Service
WF04-10	Export PDF and XLSX and distribute only to authorized recipients. Recheck each recipient's scope; generate scoped copies or deny delivery. Retain metadata in exports; log distribution.	Report Service

Figure 3.3-4 WF-04 target workflow**Figure 3.3-4 WF-04 target workflow**

WF-05 Financial Audit Review

Current-state context: Audit evidence is distributed across source systems; revenue review decisions and analytical lineage need a consolidated audit view.

Target-state steps

Step	Activity and outcome	Responsible role
WF05-01	Authenticate and choose period, organization, revenue source, invoice/case ID, status and review-decision scope.	Internal Financial Auditor
WF05-02	Retrieve the authorized population and distinct invoice/case totals without duplicate joins.	Platform
WF05-03	Assemble source snapshots, mapping evidence, validation and anomaly results, analytical changes, reviewer identity and timestamps. Preserve official source records.	Audit Service
WF05-04	Check mapping reviews, KPI changes and distribution approvals against the approved revenue responsibility and access matrix. Procurement/payment financial-delegation checks are excluded.	Audit Service
WF05-05	Report authorization deviations and proposed follow-up; route governance exceptions to the responsible Governance Officer.	Platform / Governance Officer
WF05-06	Review findings and request additional evidence or source-owner correction through existing procedures.	Internal Financial Auditor
WF05-07	Export PDF/XLSX to authorized oversight recipients, with audit ID and provenance. A digital signature, if separately approved and available, confirms integrity/authorship, not factual or legal validity. Log access independently.	Auditor / Platform

Figure 3.3-5 WF-05 target workflow

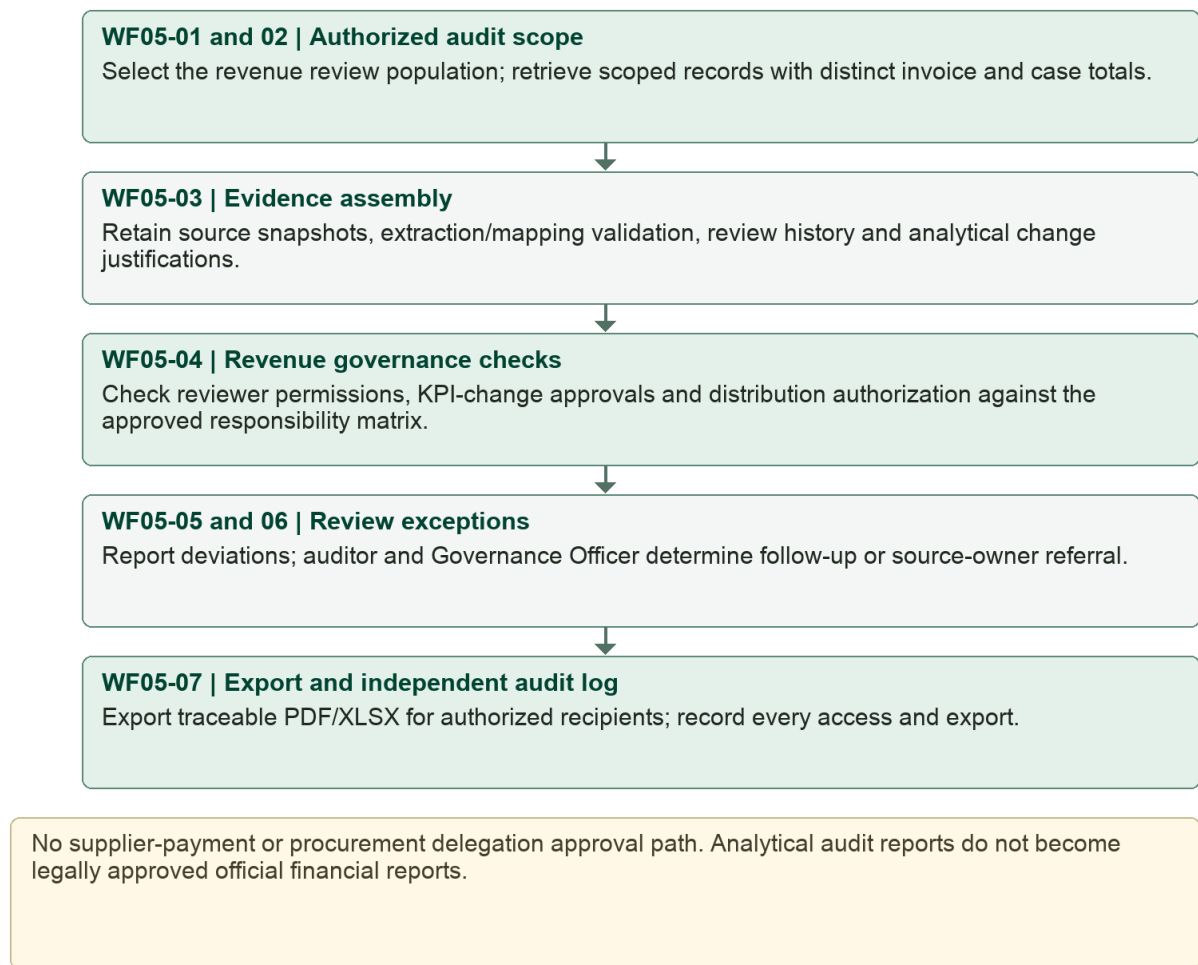


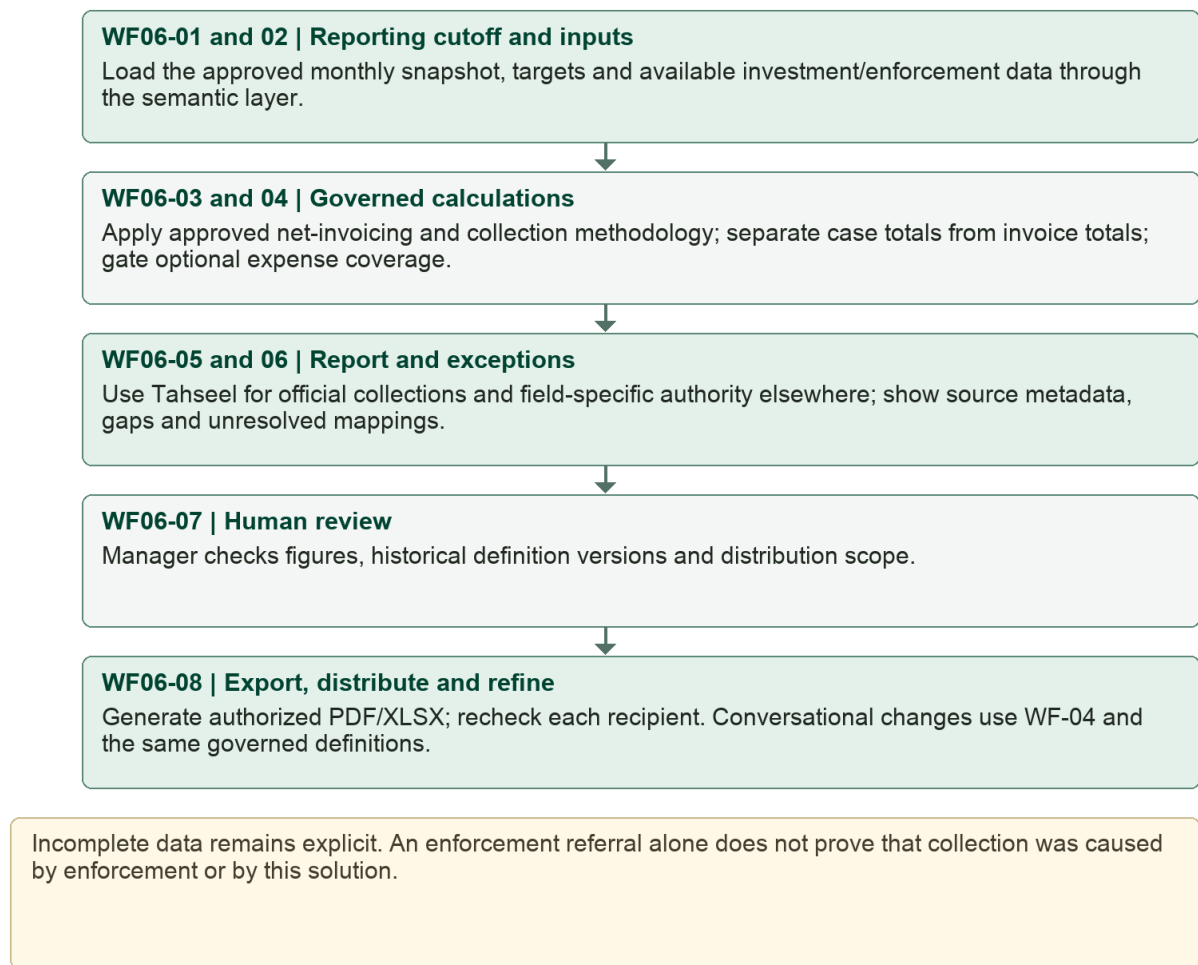
Figure 3.3-5 WF-05 target workflow

WF-06 Periodic Revenue Reporting

Current-state context: Monthly revenue reporting requires manual aggregation and validation of approved source figures and targets.

Target-state steps

Step	Activity and outcome	Responsible role
WF06-01	Run on the first working day after the approved Tahseel monthly cutoff is available. If incomplete, issue a clearly marked draft or defer distribution under the approved policy.	Scheduler
WF06-02	Load collections, invoice records, approved targets, investment/lease and enforcement mappings through the semantic layer. Expense inputs are optional and require approved finance data.	Analytics Agent
WF06-03	Calculate net invoicing, collected amount, target and year-on-year change using the approved versioned methodology and aligned period/population.	Analytics Agent
WF06-04	Calculate enforcement measures without double-counting case values or invoice balances. Show expense coverage only with an approved numerator, denominator and same-period expense dataset.	Analytics Agent
WF06-05	Compare actual collections with targets. Tahseel governs official collection figures; resolve other fields by the authority matrix and disclose unresolved differences.	Analytics Agent
WF06-06	Generate revenue-source and Amanah views with investment-revenue separation, enforcement status and metadata. Unmapped enforcement amounts remain separately disclosed; missing inputs are not zero.	Report Service
WF06-07	Review calculations, versions, variances, exceptions and recipient scope before authorizing distribution.	Revenue and Collection Manager
WF06-08	Export PDF/XLSX and execute only authorized distribution. Revalidate recipients on each scheduled run. Use WF-04 for conversational refinements to the saved report.	Report Service

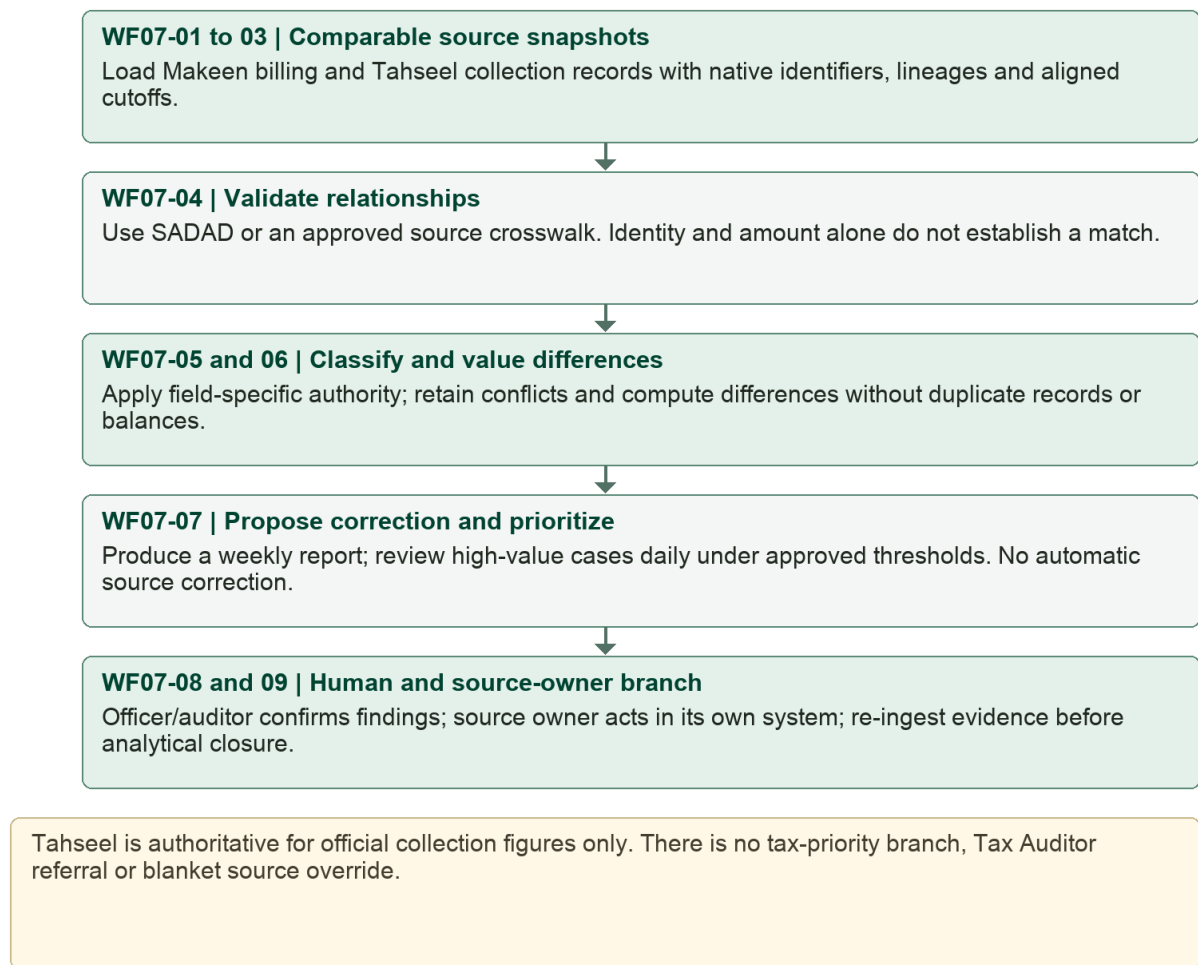
Figure 3.3-6 WF-06 target workflow**Figure 3.3-6 WF-06 target workflow**

WF-07 Financial Reconciliation

Current-state context: Makeen and Tahseel records require cross-platform matching and classification of billing and collection differences.

Target-state steps

Step	Activity and outcome	Responsible role
WF07-01	Run weekly on Sunday before 08:00 Asia/Riyadh when approved snapshots are available; show actual cutoff and reconcile comparable periods.	Platform
WF07-02	Retrieve approved Makeen billing fields, native invoice keys, any SADAD reference, amounts, status and organizational/revenue classifications; retain source lineage.	Ingestion Agent
WF07-03	Retrieve approved Tahseel SADAD references, collection amounts, collection dates and related source statuses. Authority is field-specific.	Ingestion Agent
WF07-04	Match using validated SADAD invoice numbers or approved source crosswalks. Debtor ID and amount can support investigation but cannot alone establish a relationship; unresolved candidates stay separate.	Validation Agent
WF07-05	Classify missing records, amount differences, timing differences, status conflicts and collected-versus-cancelled cases. Preserve each source value.	Validation Agent
WF07-06	Calculate each discrepancy at its defined grain and prioritize review. Tahseel governs official collection figures only; unresolved field ownership remains pending owner agreement.	Validation Agent
WF07-07	Produce proposed corrections, not source amendments. Review high-value differences daily using an owner-approved threshold; the original SAR 50,000 is a proposal awaiting confirmation.	Platform
WF07-08	Confirm the analytical reconciliation finding or refer it for audit/source-owner action. Only the authorized source-system process can amend an official record.	Reconciliation Officer
WF07-09	Review evidence and confirm referral outcome. Close the analytical issue only with evidence; verify any official correction by re-ingesting the source record.	Internal Financial Auditor

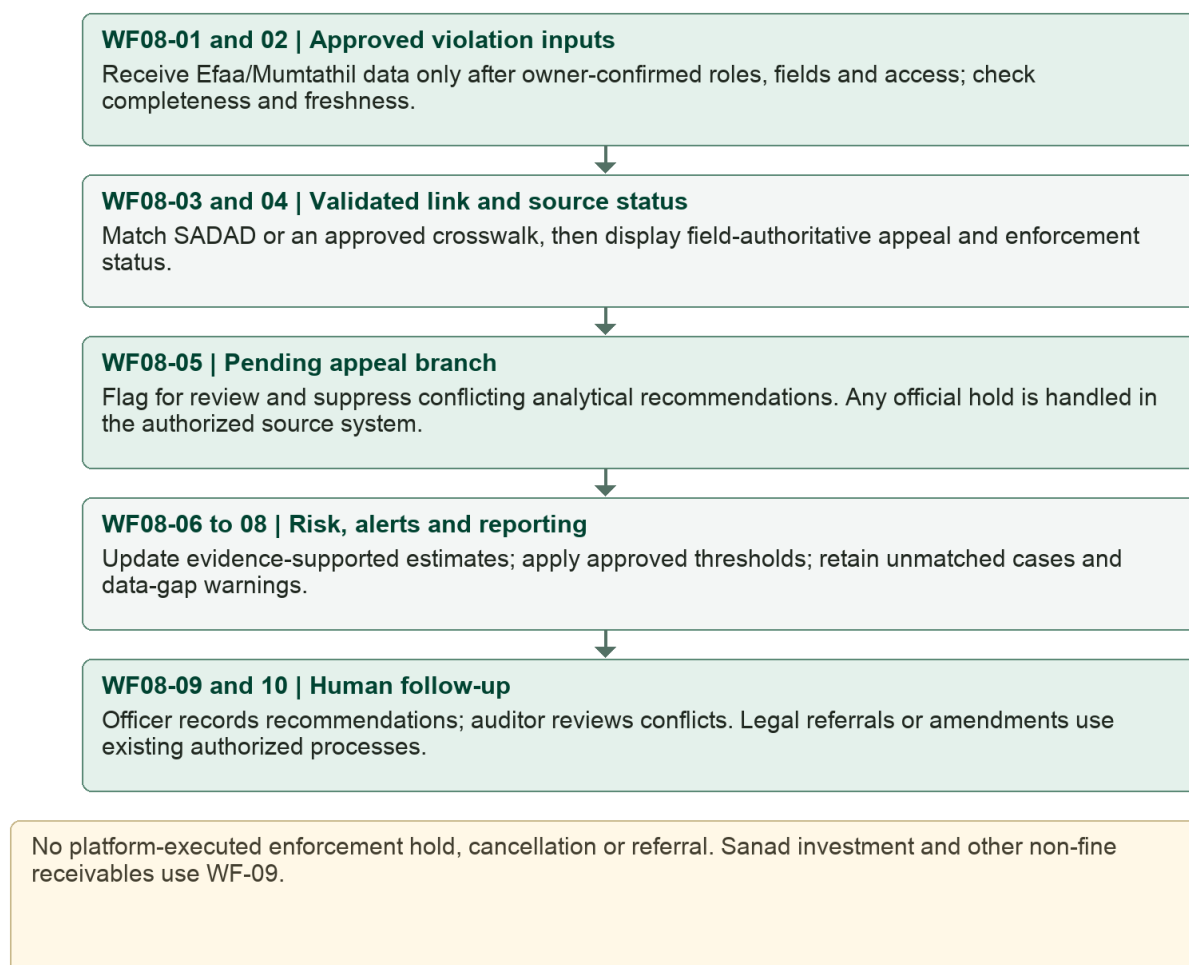
Figure 3.3-7 WF-07 target workflow**Figure 3.3-7 WF-07 target workflow**

WF-08 Violation Follow-up

Current-state context: Violation, appeal and collection status is consolidated from Efaa, Mumtathil and Tahseel; source roles and case-state ownership require data-owner confirmation.

Target-state steps

Step	Activity and outcome	Responsible role
WF08-01	Trigger on an approved violation batch or material source-status change. The provisional Efaa/Mumtathil event model must be validated with their owners before integration.	Platform
WF08-02	Validate completeness, identifiers, timestamp and source provenance; retain conflicting or stale case-state data as exceptions.	Validation Agent
WF08-03	Validate the SADAD link or approved violation-to-invoice crosswalk. Violator identity is supporting evidence only; ambiguous matches remain unresolved.	Validation Agent
WF08-04	Display source-recorded violation, appeal and enforcement status using the agreed authority matrix. Keep this fines workflow separate from Sanad non-fine receivables.	Platform
WF08-05	For a source-recorded pending appeal, flag the case for authorized review and suppress conflicting platform recommendations under approved rules. Do not execute or cancel an enforcement hold.	Platform
WF08-06	Recalculate analytical collection risk only from validated available inputs with model version and uncertainty.	Analytics Agent
WF08-07	Alert on material status or amount changes under owner-approved rules. The original below-SAR 100,000 condition is unverified and must not be activated without BRD/owner confirmation.	Analytics Agent
WF08-08	Refresh the follow-up view and weekly portfolio report through governed definitions, showing unmatched cases and source freshness.	Platform
WF08-09	Record a follow-up recommendation or referral for the authorized source-system process. The platform cannot initiate, pause, cancel or update a legal action or official violation.	Penalties and Fines Officer
WF08-10	Review identity, mapping or source-state conflicts and supporting evidence; retain unresolved cases without inferring legal outcomes.	Internal Financial Auditor

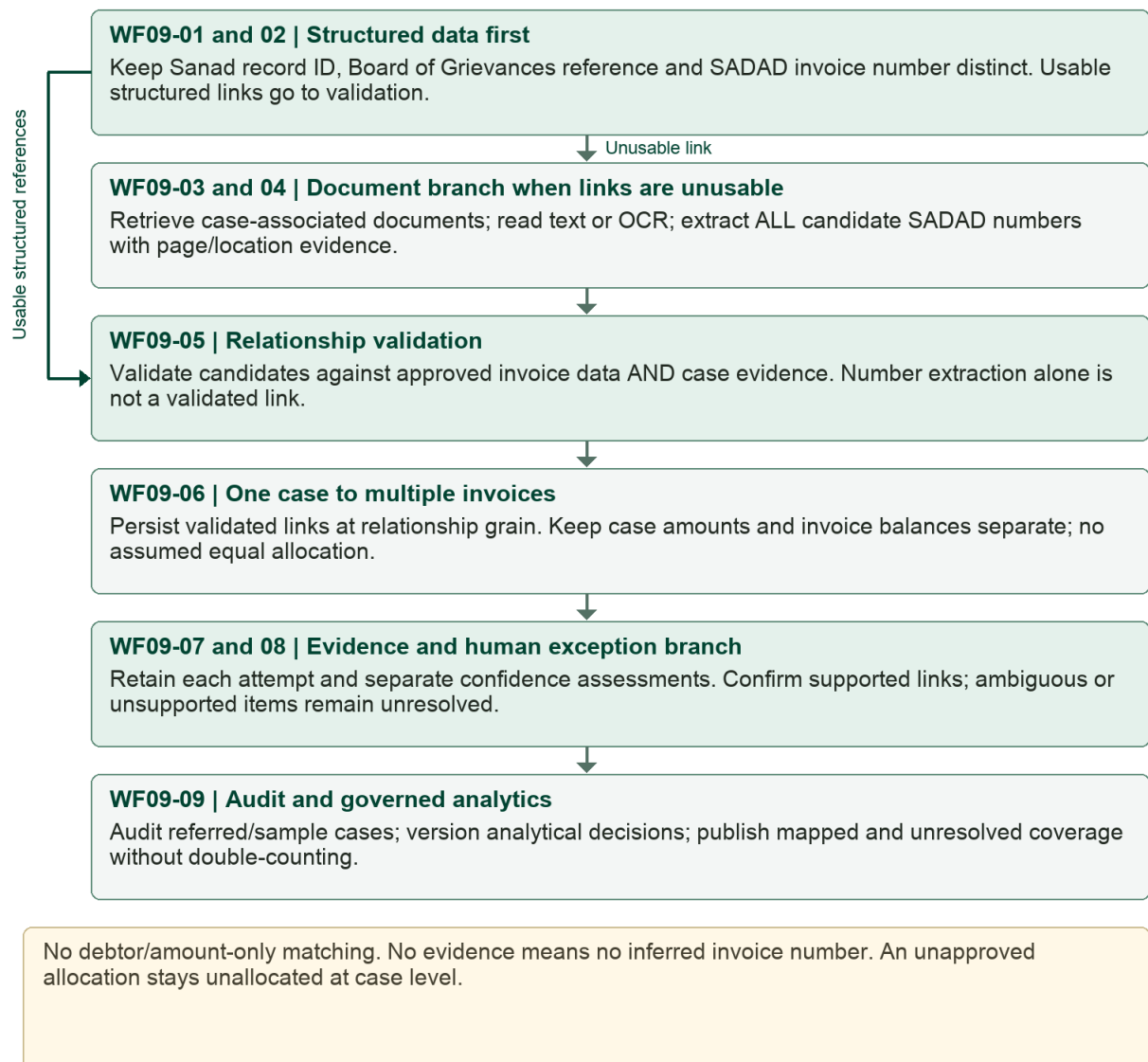
Figure 3.3-8 WF-08 target workflow**Figure 3.3-8 WF-08 target workflow**

WF-09 Sanad Enforcement-to-Invoice Mapping

Current-state context: Some Sanad records contain an enforcement reference without a usable structured link to the underlying SADAD invoices; manual document review is then needed.

Target-state steps

Step	Activity and outcome	Responsible role
WF09-01	Receive a new or updated Sanad record and its available enforcement reference. Retain Sanad record ID and Board of Grievances reference in separate fields; missing case evidence creates an unresolved exception.	Ingestion Agent
WF09-02	Inspect all structured SADAD references and validate their completeness and uniqueness. Usable references go to step 05; missing, incomplete, invalid or ambiguous references go to step 03.	Validation Agent
WF09-03	Retrieve documents demonstrably associated with that Sanad record and enforcement reference; preserve document ID, version/hash and access controls. If unavailable, remain unresolved.	Ingestion Agent
WF09-04	Read machine-readable text first; OCR only scanned/image content. Extract all candidate SADAD invoice numbers with original value, normalized value, page/location and extraction confidence. VAT is not a required mapping field.	Ingestion Agent
WF09-05	Validate structured or extracted candidates against approved invoice datasets, including Tahseel where applicable, and the case evidence. A recognized number is not proof of relationship. Never match solely on debtor identity or amount.	Validation Agent
WF09-06	Store validated case-to-invoice links separately from case values and invoice balances. One case may link to many invoices. Without an approved allocation rule, retain the amount unallocated at case level; never assume equal shares.	Validation Agent
WF09-07	Record every attempt: source record/document version, page/location where applicable, candidate value, validation evidence/result, extraction confidence, relationship assessment and reviewer/history. Structured links retain field-level provenance.	Platform
WF09-08	Review conflicts, ambiguous matches and partial cases. Confirm individual supported links, reject proposals, request evidence or retain unresolved status; show whether the case is fully or partially mapped.	Reconciliation Officer
WF09-09	Independently audit referred cases and sampled accepted mappings. Changes to analytical links are versioned; neither case nor receivable records are amended.	Internal Financial Auditor

Figure 3.3-9 WF-09 target workflow**Figure 3.3-9 WF-09 target workflow**

3.4 User Journeys

Each journey uses controlled sample data in the prototype. Users can inspect evidence and distinguish analytical state from official source state. Authorization and missing-data handling apply throughout.

ID / journey	User experience	Traceability
J-01 Intelligent Invoice Processing	Officer selects sample record → inspects structured/text/OCR evidence → reviews duplicate/validation/anomaly results → receives simulated routing → confirms or corrects the analytical proposal. A rejected proposal never cancels the receivable.	WF-01; FR-001–009/011
J-02 Collection Forecast and Follow-up	Manager selects period and portfolio → reviews horizon, uncertainty and available Furas lease features → prioritizes distinct invoices → field staff use existing collection procedures. No official status changes.	WF-02; FR-010/020
J-03 KPI and Performance Intelligence	Manager loads an approved/example catalog version → checks baseline, target and direction → explores trends and evidence → reviews recommendations/new KPI proposals → business owner approves/version-controls official changes → history remains reproducible.	WF-03; FR-017/019
J-04 Dynamic Conversational Reporting	User describes a report → clarifies ambiguity → receives an editable governed report → changes period/Amanah/investment/enforcement filters conversationally → saves a version → refreshes and exports/distributes within current recipient scope. Sources, calculations and freshness remain visible.	WF-04; FR-018/019
J-05 Financial Audit Review	Auditor scopes records → reviews mapping evidence and analytical decision history → checks revenue permissions and KPI/distribution approvals → refers findings → exports a traceable analytical audit report.	WF-05; FR-007/012
J-06 Periodic Revenue Reporting	Manager reviews a cutoff-based monthly report → verifies official collections, approved calculations and gaps → authorizes recipients → exports or requests a conversational variation using the same saved definition.	WF-06/04; FR-013/018/019
J-07 Financial Reconciliation	Officer aligns source periods → validates links → inspects differences under field authority → confirms analytical findings → refers correction to the source owner → re-ingests official evidence. Tahseel governs collections, not every field.	WF-07; FR-005/014
J-08 Violation Follow-up	Officer reviews validated violation links and source states → checks pending appeal and stale/conflict warnings → reviews analytical risk and recommended follow-up → uses the existing source process for any legal action.	WF-08; FR-015
J-09 Sanad Enforcement-to-Invoice Mapping	Officer opens a case → checks structured SADAD references → inspects associated text/OCR candidates if needed → validates every case–invoice relationship → confirms supported one-to-many links → retains partial/unresolved coverage and evidence → checks distinct case/invoice totals.	WF-09; FR-001/002/005/016/019

J-01 Detailed Prototype Journey

ID	User journey steps	Illustration
J-01.1	Start invoice intake	The Invoice and Collection Officer selects a controlled sample invoice from manual upload or a simulated source-system feed. The prototype records the source and displays a draft unique invoice identifier. No production ingestion or source-system update is performed.
J-01.2	Extract and standardize invoice data	Inspect original and normalized fields with missing-data and field-confidence indicators. Check structured references before text/OCR; Sanad linkage gaps continue under J-09.
J-01.3	Check for duplicate invoices	Compare sample source keys and invoice references; use amount/date/identity as supporting evidence only. Quarantine suspected analytical duplicates without deleting source invoices.
J-01.4	Validate receivable references	Compare extracted SADAD and relevant investment references against approved sample datasets; display relationship evidence separately from OCR confidence and route uncertainty to review.
J-01.5	Assess financial-anomaly risk	The AI capability evaluates the invoice against controlled historical patterns and presents a risk indicator, reasons, and supporting comparisons. Flagged cases are referred to the Internal Financial Auditor for review; the prototype does not automatically reject an invoice.
J-01.6	Notify responsible officer	Use a sample revenue responsibility matrix to route the case; display notification evidence. External delivery is simulated in the prototype.
J-01.7	Resolve an exception and reprocess	Review the stop reason, request evidence or correct/reject the analytical proposal with a reason; reprocess through the same checks. The original source record remains unchanged.

J-09 Mapping Review Outcomes

Observed case	Required user-visible outcome
Usable structured reference	Validate the reference and case evidence directly; document extraction is not automatically required.
Multiple SADAD numbers in documents	Show every candidate with page/location evidence and independent validation state.
Some supported links and some missing evidence	Show a partially mapped case, supported links and unresolved remainder; no fabricated invoice or equal allocation.
No supported relationship	Keep unresolved; request evidence. Extracted numbers alone do not establish a relationship.
Confirmed mapping	Version the analytical links and show distinct invoice totals, case totals and any unallocated amount separately.

4 Requirements Summary

This revised HLSD contains the requirement and acceptance baseline for review. DOC-01 Product Requirements and DOC-02 Service Mapping must be synchronized by their owners before submission; their contents or approval status cannot be verified from the supplied files.

4.1 Business Requirements

ID / requirement	Required behavior	Accountable owner
FR-001 Controlled multi-source intake	Ingest approved structured invoice/receivable datasets and Sanad records with their associated enforcement documents. Check structured SADAD links before document retrieval; preserve native IDs and provenance. Support PDF, DOCX, XML and XLSX through approved channels; proposed 50 MB/file limit, with validation and quarantine. Connectors are not assumed to exist.	Invoice and Collection Officer P0
FR-002 SADAD extraction and evidence	Read structured data and machine-readable text before OCR; support Arabic, English and mixed-language material. Extract every relevant SADAD invoice number and retain document/page evidence and field confidence. VAT is optional source data, not a Sanad objective; unconfirmed payment-order fields are excluded.	Invoice and Collection Officer P0
FR-003 Duplicate detection	Detect duplicate source keys, repeated files and candidate invoice duplicates without merging solely by amount or identity. Quarantine analytical duplicates for review; preserve official source records and replay idempotently.	Invoice and Collection Officer P0
FR-004 Normalization and provenance	Preserve original and normalized identifiers, leading zeros, Arabic/Latin digits, currencies, dates and source versions. Distinguish analytical IDs from Sanad IDs, Board of Grievances references and SADAD numbers; record every transformation.	Data Lead P0
FR-005 Receivable relationship validation	Validate invoice/SADAD and investment-receivable relationships using approved datasets and crosswalks. Store evidence and match outcomes separately from extraction confidence; identity and amount alone are insufficient. Route unsupported matches to unresolved review.	Financial Reconciliation Officer P0
FR-006 Amount consistency and data quality	Check available components, totals, adjustments and balances against approved arithmetic rules. Retain relevant source VAT only as optional amount context; distinguish missing components from zero. Exclude tax assessment, registration checks, VAT compliance and all ZATCA integration.	Internal Financial Auditor P0
FR-007 Revenue review responsibilities	Maintain owner-approved routing and review responsibilities by organization, revenue domain and risk. Version rules and effective dates; unconfirmed thresholds remain disabled. Exclude procurement and supplier-payment financial delegation.	Governance Officer P0
FR-008 Revenue anomaly detection	Flag unusual amounts, duplicate receivables, inconsistent statuses and revenue patterns with evidence and calibrated risk scores; support human dispositions and model monitoring. Findings are indicators, not determinations of fraud.	Revenue and Collection Manager P0
FR-009 Controlled review routing	Route cases under FR-007 and log authorized notifications, ownership and escalation. The prototype simulates delivery; production notifications require approved channels. No payment authorization or legal action is executed.	Revenue and Collection Manager P0
FR-010 Collection forecasts and analytics	Use governed, authorized invoice, collection, investment and enforcement features for horizon-defined collection/risk estimates and dashboards. Disclose versions, uncertainty, freshness and unavailable outcomes; apply FR-017 for KPIs and FR-018 for conversation/reporting.	Revenue and Collection Manager P0

ID / requirement	Required behavior	Accountable owner
FR-011 Analytical exception resolution	Allow authorized review, correction/reprocessing of derived proposals, evidence requests, rejection and escalation. Preserve source records and history. Rejecting an extraction, classification or mapping never cancels a municipal receivable.	Invoice and Collection Officer P0
FR-012 Revenue audit and governance	Search immutable/tamper-evident audit evidence and check mapping-review, KPI-change and distribution permissions. Export scoped PDF/XLSX with provenance; optional approved digital signatures prove integrity/authorship, not official financial validity.	Internal Financial Auditor P0
FR-013 Periodic revenue reports	Generate monthly source/Amanah reports with investment-revenue separation, target comparisons and deduplicated enforcement analysis. Expense coverage requires approved expense inputs/formula. Reuse FR-018 definitions and controlled distribution; include freshness and gaps.	Revenue and Collection Manager P0
FR-014 Field-authoritative reconciliation	Reconcile Makeen/Tahseel data using approved links and comparable cutoffs. Tahseel governs official collection figures only. Classify other differences by the agreed authority matrix; retain unresolved conflicts and propose corrections for the source owner.	Financial Reconciliation Officer P0
FR-015 Violation status intelligence	Consolidate owner-confirmed Efaa/Mumtathil and Tahseel data; validate invoice links, display authoritative appeal/enforcement states and recommend follow-up. The platform cannot initiate, hold, cancel or amend a legal action or violation.	Penalties and Fines Officer P0
FR-016 Sanad enforcement-to-invoice mapping	Implement WF-09: validate structured references first; retrieve documents for unusable links; extract all SADAD candidates; validate case relationships; retain distinct IDs, evidence and separate confidence assessments. Preserve one-to-many links and unresolved cases. No equal allocation without approval; no double-counting.	Financial Reconciliation Officer P0
FR-017 KPI and Performance Intelligence	Consume approved formulas, owners, periods, baselines, targets and improvement direction. Specify and approve baseline period/population/method. Explain trends/deviations with evidence, recommend follow-up and propose new KPIs. Official changes require approval, version/effective dates and preserved historical commitments.	Business KPI Owner P0
FR-018 Dynamic conversational reports	Create editable reports from approved measures, dimensions, calculations, filters, periods, comparisons and charts. Clarify ambiguity, disclose unavailable data, support conversational refinement, versioned saves/refreshes, PDF/XLSX and controlled recipient-scoped distribution. Every output retains source/definition/freshness metadata.	Shared Services Leader P0
FR-019 Governed semantic metrics service	Maintain reusable entities, source authority, relationships, status/date meanings, organizational/revenue classifications and approved measures. Encode net-invoicing and collection-rate numerator, denominator, exclusions and time alignment as approved versioned rules. Prevent raw-table bypass and duplicate enforcement joins.	Data Governance Lead P0
FR-020 Investment and lease revenue analysis	Ingest approved Furas-originated contract data directly, through a verified existing integration, or via an approved dataset. Link investment/lease terms to invoices and collections for analysis, matching, follow-up and forecasting. Keep procurement and contract-award/amendment administration excluded.	Revenue and Collection Manager P0

P0 = core required capability. Requirements are accountable to human business/data owners; agents are implementing services, not approvers. FR-020 gives the Furas use case an explicit requirement ID and treats it as core scope; access blockers must be escalated, not silently downgraded.

4.2 Non Functional Requirements

Numerical targets below are proposals carried forward or clarified for approval. They are not represented as measured results or signed service commitments.

ID / category	Requirement
NFR-001 Performance	Proposed target: P95 ≤3 seconds for navigation, status and bounded search at 50 concurrent sessions. Complex report generation, OCR and batch jobs are asynchronous with progress/status; their workload-specific SLAs require load-test approval.
NFR-002 Access control	Enforce identity, role, organization and row/column scope at query, evidence retrieval, report save, refresh, export and recipient distribution. Test revoked access, cross-Amanah access and indirect data leakage; log denied attempts.
NFR-003 Availability and recovery	Proposed ≥99.5% availability in the approved 07:00–22:00 working-day calendar, measured by service probes. Specify maintenance treatment, failover, daily encrypted backup and restore testing; RPO/RTO and calendar require IT approval before Beta exit.
NFR-004 Audit and retention	Record source snapshots, mapping attempts, confidence, reviewer decisions, definition/model/report versions, access and distribution. Use tamper-evident append-only storage and controlled access. The source's seven-year retention is proposed, subject to records/privacy-owner approval before production.
NFR-005 Data quality	Validate schema, currency, date semantics, mandatory fields and source identities; quarantine defects. Do not manufacture missing SADAD references to meet completeness targets. Report unresolved mapping coverage separately from source-data completeness.
NFR-006 Capacity	Proposed peak 500 structured invoice records/minute and 3× agreed baseline volume; establish baseline workload, file-size mix and hardware first. OCR pages and report jobs need separate throughput measurements. Validate scaling and backpressure without dropped/duplicated records.
NFR-007 Controlled maintenance	Version and approve rule/configuration changes with rollback; use staged releases and backward-compatible migrations. Zero-downtime deployment is a design target subject to architecture validation, not an unconditional guarantee.
NFR-008 Security and data protection	Proposed OCI deployment must satisfy approved Saudi residency, government-network and applicable program security/privacy controls. Encrypt stored data with approved AES-256 configuration and traffic with TLS; approve model/OCR endpoints, key management and document isolation before real data use.
NFR-009 AI evaluation	Use task-specific validation in Section 5.4: SADAD exact extraction, relationship precision/recall, anomaly precision/recall and forecast calibration/error by horizon. Avoid one blanket 85% accuracy requirement. Evaluate representative held-out data, drift and quarterly model reviews.
NFR-010 Integration and freshness	Proposed ≥99% successful eligible requests after bounded retries, with raw failures and retry outcomes reported separately. Use source checkpoints, idempotency, circuit breakers and stale-data warnings. Agree source cadence and freshness SLAs before production; do not assume live APIs.
NFR-011 Metrics governance	Only authorized owners may approve/version definitions and effective dates. KPI, forecasting and reporting services consume governed measures/features; block conversational direct access to raw source tables. Pin historical definitions and prevent silent report-definition migration.
NFR-012 AI query and evidence safety	Treat source documents and prompts as untrusted input. Validate report specifications against an allowlisted schema, use read-only parameterized queries, impose resource limits and restrict evidence to user scope. Test prompt injection, invented formulas and unauthorized exports.

5 Success Metrics and Acceptance

Ministry business KPIs managed in WF-03 are separate from the project-benefit and technical measures in this section. The supplied files contain no approved KPI catalog or evidence that the numerical project targets have been signed off. All targets below are proposed; the Solution Owner and relevant data/IT owners must approve baselines, definitions and acceptance thresholds.

5.1 Strategic Benefit Metrics

Released capacity: the source assumes 1,632 hours/FTE/year and SAR 270,000/FTE/year. Its 6M/Y1/Y2 scenarios of 4/9/15 FTE imply 6,528/14,688/24,480 annualized hours and SAR 1.08M/2.43M/4.05M annualized capacity value. These are run-rate scenarios at each milestone, not cumulative savings. A 6-month result must not be counted as a full year of realized savings.

The source's SAR 2.70M/year baseline equals ten FTE at the same rate and does not substantiate a fifteen-FTE release. Remove it as an approved baseline until the eligible workforce and workload are verified. Measure net time saved from comparable pre/post task samples, subtract new review/maintenance effort, and cap claimed capacity at observed eligible hours. Redeployment alone is not a cash saving; a cash benefit requires verified avoided/reduced expenditure, net of implementation and operating cost.

Measure	Proposed assessment	Owner / cadence
Released employee capacity	Baseline: measured eligible task hours, not yet established. Indicative run-rate scenarios above require workload validation. Report hours and FTE equivalents separately from money.	Solution Owner / Finance; monthly observations, annual validation
Solution-attributable collections	Measure incremental Tahseel-confirmed collections in a treated, eligible invoice cohort versus a comparable control or approved counterfactual, adjusted for mix, policy and timing. Prevent duplicate invoices and acceleration-versus-lifetime uplift confusion.	Revenue Owner / Finance; monthly cohorts, annual assessment
Benefit commitment gate	Approve baseline period, population, assumptions, cost inputs, attribution method and uncertainty before using targets as commitments.	Solution Owner / Finance; before release benefits baseline

Revenue growth: v0.3 assumes SAR 16.7B for 2026, 3.1% underlying GDP growth and 4.65% collection growth, yielding roughly SAR 17.5B and SAR 18.3B. These are unverified planning inputs, not a demonstrated causal benefit. GDP growth is not accepted as a collection counterfactual. Withdraw these amounts as solution targets until Finance approves a suitable collection baseline and attribution design. Report gross revenue trends separately from solution-attributable collections.

5.2 Operational Benefit Metrics

Measure	Definition and proposed target	Measurement / owner
Employee time saved	Net hours released from comparable tasks; milestone annualized scenarios: 6,528 / 14,688 / 24,480, subject to validated eligible workload.	Task time studies with volume/complexity controls and review effort deducted; Solution Owner, monthly.
Employee adoption	Monthly active authorized users completing a meaningful governed task ÷ approved eligible user population. Proposed 6M/Y1/Y2: 50% / 80% / 100%.	Report population changes, exclusions and usage events; Solution Owner, monthly.

Measure	Definition and proposed target	Measurement / owner
Processing/ reconciliation time	Median and P90 elapsed working time per comparable completed case; baseline period/population to be approved. No committed 50% reduction without evidence.	Pre/post matched workload samples; Revenue Owner / Operations, monthly.

5.3 Technical Service Measures

Measure / requirement	Proposed target and method	Owner / cadence
Availability NFR-003	≥99.5% within approved service calendar, measured by service probes as available minutes ÷ eligible minutes. Record exclusions and incidents; use restore drills for recovery evidence.	IT / monthly
Interactive latency NFR-001	P95 ≤3 seconds at 50 concurrent sessions for bounded interactive operations; measure complex reports/OCR separately under approved job SLAs.	IT / release and monthly
Structured throughput NFR-006	500 records/minute and 3× approved baseline; load tests specify payload size, duration, error rate, resources and scaling. OCR pages/minute tested separately.	IT / release and quarterly
Source-data quality NFR-005	Propose ≥99% completeness for owner-designated required fields and ≤0.1% invalid values on eligible input fields; missing SADAD mappings remain unresolved and are measured separately.	Data Lead / weekly
Integration reliability NFR-010	Propose ≥99% eligible requests successful after bounded retries; also publish first-attempt success, failures and freshness. Use actual monitored request counts.	Integration Lead / monthly
Access and audit NFR-002/004	100% required access/approval events logged and negative authorization tests passed; audit the approved retention configuration and restoreability. Test results do not guarantee absence of future incidents.	Security / release and quarterly
AI quality NFR-009	Use AC-01/02/10 task-specific metrics and representative held-out data. No blanket “85% across AI tasks” acceptance.	AI Lead + business owners / release and quarterly

5.4 Capability Acceptance Tests

The following are proposed acceptance criteria for owner agreement, not test results. Before evaluation, approve a representative, access-controlled gold dataset with sufficient cases per risk stratum and independent adjudication. Record sample sizes, prevalence, confidence intervals and coverage; keep model-tuning data separate from held-out acceptance data. Report precision as correct accepted items divided by accepted items, and recall as recovered gold items divided by eligible gold items.

Prototype checks demonstrate behavior with samples and seeded exceptions; they cannot establish production accuracy. Alpha verifies components; Beta evaluates approved integrations, formulas and representative data. Production release requires signed acceptance, resolved critical defects and the gates in Section 6.3. Exact deterministic controls use 100% pass criteria within the agreed suite, not claims of universal error-free operation.

AC-01 SADAD extraction

Requirements: FR-001/002/004/016. Propose $\geq 99\%$ exact-token precision and $\geq 95\%$ recall across all candidate SADAD numbers, with leading zeros preserved; report Arabic/English, text/scanned and multi-invoice strata separately.

Validation evidence: Gold-labeled documents with correct page locations; include multiple numbers, non-SADAD identifiers, missing/illegible scans and structured-reference bypass cases.

AC-02 Validated relationships

Requirements: FR-005/016. Propose $\geq 99\%$ precision for accepted links and $\geq 95\%$ recall on resolvable gold links; report automated and human-confirmed results separately. Require zero unsupported auto-acceptances in the negative test suite.

Validation evidence: Independent owner adjudication of case–invoice links; include a valid-looking number belonging to a different case, conflicting sources and debtor/amount-only candidates.

AC-03 One-to-many and unresolved cases

Requirements: FR-014/016/019. 100% of deterministic test cases preserve supported links and unresolved/partial status; no inferred invoice numbers, duplicate balances, repeated case values or unapproved equal allocations.

Validation evidence: One case with several invoices; duplicate documents; partial mapping; no evidence; one invoice linked to multiple cases; unallocated case amount. Reconcile totals to hand-checked gold results.

AC-04 Mapping evidence

Requirements: FR-002/012/016. 100% of mapping attempts have original/normalized candidate, source record/version, document/page where applicable, validation result, extraction confidence, relationship assessment and review history.

Validation evidence: Inspect structured and OCR attempts, rejected links and reprocessed documents; verify reviewers can retrieve only evidence within their access scope.

AC-05 Reports and semantic calculations

Requirements: FR-010/013/018/019. 100% exact agreement with approved expected results at the defined currency/rounding precision; zero duplicate totals or invented values in deterministic tests.

Validation evidence: Net invoicing, collection numerator/denominator/exclusions, zero denominator, reversals, period alignment, investment filters, Amanah comparison and enforcement partitions; verify export metadata.

AC-06 Access and controlled distribution

Requirements: FR-012/018; NFR-002/011/012. 100% of specified negative tests deny unauthorized queries, evidence, refreshes and exports; zero data leakage in the test suite.

Validation evidence: Cross-Amanah and column restrictions, revoked users, scheduled jobs, narrower-scope recipients, shared links and prompt-injection attempts.

AC-07 KPI governance and history

Requirements: FR-017/019; NFR-011. 100% of official definition/baseline/target/threshold changes require authorized approval, effective date and immutable version history; historical views reproduce prior commitments.

Validation evidence: Increase/decrease/range KPIs; baseline period/population/method; rejected proposal; future-effective change; restatement; analytical recalibration without changing official targets.

AC-08 Conversational report lifecycle

Requirements: FR-013/018. 100% pass on the agreed scenario suite: editable creation, refinement, save, refresh, PDF/XLSX export, recipient controls, clarification and missing-data disclosure.

Validation evidence: Change period, compare Amanahs, separate investment revenue and enforcement referrals; re-open saved definition and verify version/policy compatibility.

AC-09 Source authority and action boundaries

Requirements: FR-006/007/011/014/015/020. 100% of boundary tests retain conflicting source values, follow field authority and prevent source writes, payment approvals, tax checks or platform-executed legal actions.

Validation evidence: Tahseel collection versus Makeen billing conflict; pending appeal; rejected mapping; daily high-value review; owner-authorized correction re-ingested.

AC-10 Forecast and anomaly usefulness

Requirements: FR-008/010; NFR-009. Proposed classification F1 ≥ 0.85 only for approved labeled outcomes; report precision/recall and class prevalence. Forecast probabilities require calibration/Brier score and monetary forecasts error by horizon versus an agreed non-AI baseline. Final thresholds need owner approval.

Validation evidence: Time-based held-out data without leakage; segment by Amanah/revenue source and include cold-start debtors. Do not report a single “accuracy” across dissimilar tasks.

AC-11 Integration and operations

Requirements: FR-001/004; NFR-001/003/004/006/010. Validate the proposed performance, availability and reliability targets in Section 5.3 against an agreed load profile; 100% pass for replay/idempotency and restore evidence in the scenario suite.

Validation evidence: Delayed feeds, unavailable documents, retries, duplicate batches, peak structured throughput, separate OCR workload, source freshness, rollback and restore drills.

AC-12 Investment data readiness

Requirements: FR-020. All required Furas fields and feed route are owner-confirmed or explicitly unavailable; sample lease terms link to the correct invoice without implying procurement administration.

Validation evidence: Contract/version IDs, investor and asset references, schedule/amounts, approved invoice crosswalk and provenance. Exercise direct/verified-intermediary/approved-dataset route selected by owner.

6 Roadmap and Phasing

6.1 High Level Delivery Roadmap

Original v0.3 milestone dates are retained below for traceability only. As of this revision, prototype completion, production integrations and approvals have not been verified. The Delivery Lead must confirm feasibility and re-baseline dates affected by the clarified scope, access or schedule constraints; this document does not certify the 15 September prototype milestone as complete.

Phase	Required milestone evidence	Original proposed start	Original proposed end
Prototype	Sample/simulated end-to-end journeys and diagrams reviewed; updated requirements/service mapping and open-decision register available.	06/07/2026	15/09/2026
Alpha	Approved technical design, source/field contracts, semantic model and component tests; record unresolved access blockers.	16/09/2026	15/10/2026
Beta	Integrated staging validation against representative data, approved formulas, permissions and acceptance criteria.	16/10/2026	07/12/2026
Release & Stabilization phase	Signed release readiness, production access/security/operations approval, deployment and stabilization evidence.	08/12/2026	21/01/2027
Operate phase	Operational handover, monitoring, KPI tracking and ongoing solution support	After Go-Live	Ongoing

6.2 Capability Delivery Traceability

P0 identifies core business scope, not a guarantee that every production feed is available in the prototype. Owners may approve phased staging, but material deferrals require explicit scope/schedule change control. AC references point to Section 5.4; functional owner assignments are proposed until confirmed.

Capability and requirements	Accountable owner / priority	Dependencies	Delivery and acceptance
Intake, normalization, duplicate controls FR-001–004	Data Lead + Invoice Officer P0	Approved source schemas/access; representative documents; document security	Prototype: simulated intake. Alpha: controlled adapters. Beta: integrated replay/load tests. Production: source approval. AC-01/04/11.
Sanad mapping and validation FR-005/016	Reconciliation Officer P0	Sanad documents; approved invoice data; case evidence; allocation policy only if allocation is needed	Prototype: structured/text/OCR, multi-invoice and unresolved examples. Alpha: evidence model. Beta: adjudicated integration tests. Production: source/owner gates. AC-01–04.
Revenue review and exceptions FR-006–009/011/012	Revenue Manager + Governance Officer P0	Revenue responsibility matrix; risk rules; audit/retention policies	Prototype: simulated routing. Alpha: permission/audit controls. Beta: review and notification tests. Production: authorized channels. AC-04/06/09/10.
Forecasting FR-010	Revenue Manager + AI Lead P0	Historical labeled data; approved horizons/features; semantic layer	Prototype: illustrative scores. Alpha: baseline model. Beta: held-out validation. Production: approved model monitoring. AC-03/05/10.

Capability and requirements	Accountable owner / priority	Dependencies	Delivery and acceptance
KPI intelligence FR-017	Business KPI Owner P0	Approved catalog, baseline method/period/population, targets and directions	Prototype: sample versioned catalog. Alpha: governance service. Beta: approved formulas/history tests. Production: owner sign-off. AC-05/07.
Dynamic and periodic reports FR-013/018	Shared Services Leader P0	Semantic layer; approved queries/model endpoints; recipients; optional expense dataset	Prototype: editable sample reports and simulated refresh/export. Alpha: report engine. Beta: refresh/distribution integration. Production: access gates. AC-05/06/08.
Field authority and reconciliation FR-014	Reconciliation Officer + Data Owners P0	Makeen/Tahseel data contracts, agreed field authority and matching rules	Prototype: seeded conflicts. Alpha: authority registry. Beta: source snapshots. Production: reconciliation approval. AC-03/05/09/11.
Violation follow-up FR-015	Penalties and Fines Officer P0	Efaa/Mumtathil owner-confirmed roles and source statuses; invoice crosswalk	Prototype: simulated appeal/conflict cases. Alpha: agreed event model. Beta: read-only integrated tests. Production: owner approval. AC-06/09/11.
Semantic metrics architecture FR-019; NFR-011/012	Data Governance Lead P0	Approved measures, master data, authorization and authority registry	Prototype: core entities and illustrative calculations. Alpha: governed service. Beta: approved formula/access tests. Production: catalog sign-off. AC-03/05-07.
Furas investment/lease analysis FR-020	Revenue Manager + Furas Owner P0 core scope	Confirmed fields, feed route, permitted access and invoice crosswalk	Prototype: illustrative lease data. Alpha: approve route/schema. Beta: validate approved feed. Production: owner sign-off; blockers require explicit re-baselining. AC-05/12.

6.3 Open Decisions and Release Gates

ID	Open decision or dependency	Accountable owner and gate
D-01	Validate Makeen/Efaa/Mumtathil descriptions and agree material-field authority; confirm Sanad legal-state provenance.	Relevant data owners + Data Governance Lead; before related Alpha adapters are baselined.
D-02	Approve Furas minimum dataset and direct/verified-intermediary/approved-dataset route; no integration is assumed.	Furas owner + integration lead; Alpha design gate.
D-03	Provide approved net-invoicing/collection-rate formulas and full KPI catalog, baseline methods, targets and effective versions.	Business KPI Owner; before official KPI calculations or Beta acceptance.
D-04	Approve revenue responsibility matrix, alert thresholds and any enforcement amount-allocation policy. Original SAR 50,000 and below-SAR 100,000 rules are unconfirmed.	Revenue/Governance owners; before rule activation. No allocation policy means case amounts remain unallocated.
D-05	Approve representative labeled validation data, numerical thresholds, workload profile, freshness SLAs and release criteria.	Business owners + QA/IT/AI leads; before acceptance testing and Beta exit.
D-06	Validate OCI/network design, approved OCR/model services, residency, retention, RPO/RTO, key management and recipients.	Architecture, Security, Records and IT owners; before real data or production deployment as applicable.
D-07	Validate benefit baselines/attribution; confirm optional finance expense data and any claimed cash saving.	Solution Owner + Finance; before benefit commitments and expense-coverage publication.
D-08	Update DOC-01 product requirements and DOC-02 service mapping, resolve UC/screen identifiers, attach approved versions and record reviewer confirmation before Manafeh. Neither companion file was supplied.	Document owner + Solution Owner; submission gate.
D-09	Confirm revised-package completion and feasibility of all original milestone dates. Record a re-baseline if access or approvals delay delivery.	Delivery Lead + Solution Owner; current planning gate.

Additional Resources

ID / document	Status and required update
DOC-01 Product Requirements	Not supplied with this request; no usable link or embedded file exists in source v0.3. Owner must synchronize FR-001–020, NFR-001–012, AC-01–12, workflow/journey IDs and approved business rules, then attach the version/date/link.
DOC-02 Solution to Services Mapping	Not supplied with this request; source has an unfilled attachment/link field. Owner must map revenue/receivables capabilities to confirmed MOMAH services, source owners and interfaces, preserve procurement/tax exclusions and attach the approved version.

The uploaded department feedback also requests a tracked-changes copy, completion-date confirmation and final reviewer approval before Manafeh submission. The current request is fulfilled with a clean revised HLSD and a disposition matrix. Companion-document synchronization, tracked-change packaging for the broader submission, date confirmation and reviewer sign-off remain owner actions; no delivery to recipients or Manafeh submission is represented as performed.

Appendix

A Source References

Source	Exact supplied file and use
S-01	Intelligent-Invoice-Management-HLSD-V0.3.docx — uploaded design dated 14/09/2026; baseline content, NHCI identity, workflow and requirement IDs, original planning assumptions.
S-02	粘贴的文本 (1).txt — uploaded department feedback; numbered comments 1–9 and submission instructions. Its page references refer to v0.2, so this revision uses section/workflow/requirement IDs.

No external platform description, legal interpretation, KPI approval or source-system integration was treated as verified evidence. Working descriptions and proposed controls require the owners identified in Section 6.3.

B Approval Sign Off

Role	Name	Signature	Date	Supporting document
Solution Vendor (Prepared By)	ByteDance			Revised HLSD; approval pending
NHCI Oversight (Reviewed By)				
Solution Owner (Reviewed By)				
AI Champion (Reviewed By)				
Central AI (Reviewed By)				
Sector Head (Approved By)				
AI Solutions Steering Group (Approved By)				
Central AI Committee (Approved By)				

Supporting evidence may include meeting minutes recording review or approval. Blank sign-off fields are intentional and indicate pending approval.